

(1) Murphy 6.47 Assume the fuel is mixed with air at 25°C and react adiabatically. Calculate the outlet temperature for the products to estimate the flame temperature.

(2) Murphy 6.48

(3) A fuel gas containing 85 mol% methane and the balance ethane is burned completely with pure oxygen at 25°C, and the products are brought back down to 25°C.

(a) Suppose the reactor is continuous. Take a basis of 1 mol/s of fuel gas and combust with more than sufficient oxygen to calculate the heat rate transferred from the reactor per mole of fuel fed to the reactor  $\hat{Q} = \dot{Q}/\dot{n}$  in kJ/mol.

(b) Rework the problem for a constant-volume batch reactor. Use a basis of 1 mol of the fuel initially charged into the reactor and calculate the heat transfer per mole of fuel charged in kJ/mol.

(c) Briefly explain why the results in parts (a) and (b) do not depend on the percent excess oxygen.

(4) Murphy 6.68